RAJALAKSHMI
ENGINEERING COLLEGE
An AUTONOMOUS Institution
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AI-Based Cyber Defence Allocation Using Game Theory

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OUTLINE

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ABSTRACT

Organizations today face constant cyber threats but have only limited resources to fully secure every digital asset. Deciding which assets to protect first is a high-stakes decision, since the wrong choice can leave high-value systems such as payment gateways or customer databases exposed to attackers. This project proposes an AI-based Cybersecurity Resource Allocation model that treats defense planning as a two-player, zero-sum adversarial game between a Defender and an Attacker.

The system applies the Minimax algorithm, an Artificial Intelligence adversarial search technique, to compute the Defender's optimal allocation of a limited protection budget. Alpha-Beta pruning is applied to reach the same optimal decision while exploring far fewer possibilities, and a heuristic evaluation function allows the search to scale to larger, more realistic scenarios. Based on the chosen assets and budget, the system efficiently computes and explains the allocation that minimizes worst-case damage from a rational attacker.

OBJECTIVE

- To develop an AI-based cybersecurity resource allocation system.
- To implement the Minimax algorithm to compute the Defender’s optimal allocation strategy.
- To implement Alpha-Beta pruning and demonstrate its efficiency over plain Minimax.
- To design a heuristic evaluation function so the search scales to larger, realistic scenarios.
- To build an interactive system where a user can play as Defender or Attacker against the AI.
- To provide a transparent, explainable method for prioritizing limited security resources.

Literature Survey

TITLE	YEAR	AUTHOR	TECHNIQUE
Artificial Intelligence: A Modern Approach (Minimax & Alpha-Beta Search)	2021	S. Russell, P. Norvig	Minimax & Alpha-Beta Pruning
Network Security: A Decision and Game-Theoretic Approach	2010	T. Alpcan, T. Başar	Game-Theoretic Modeling
A Survey of Game Theory as Applied to Network Security	2010	S. Roy et al.	Game Theory Survey / Resource Allocation

EXISTING SYSTEM

- The existing approach relies on manual risk assessment by security teams, using **checklists and static, rule-based scanners such as** CVSS-based scoring **tools** to decide which assets to protect.
- Relies on human judgment rather than a systematic decision method.
- Ignores an intelligent, adaptive attacker.
- Scores each asset independently instead of the full allocation trade-off.
- Assessments are inconsistent between different security assessors.
- Does not guarantee optimal use of a limited security budget.
- Cannot quantify the worst-case damage under a given allocation.

PROPOSED SYSTEM

- The proposed system models Defender-vs-Attacker resource allocation as a two-player game, using the *Minimax and Alpha-Beta Search** to calculate the optimal defense allocation.

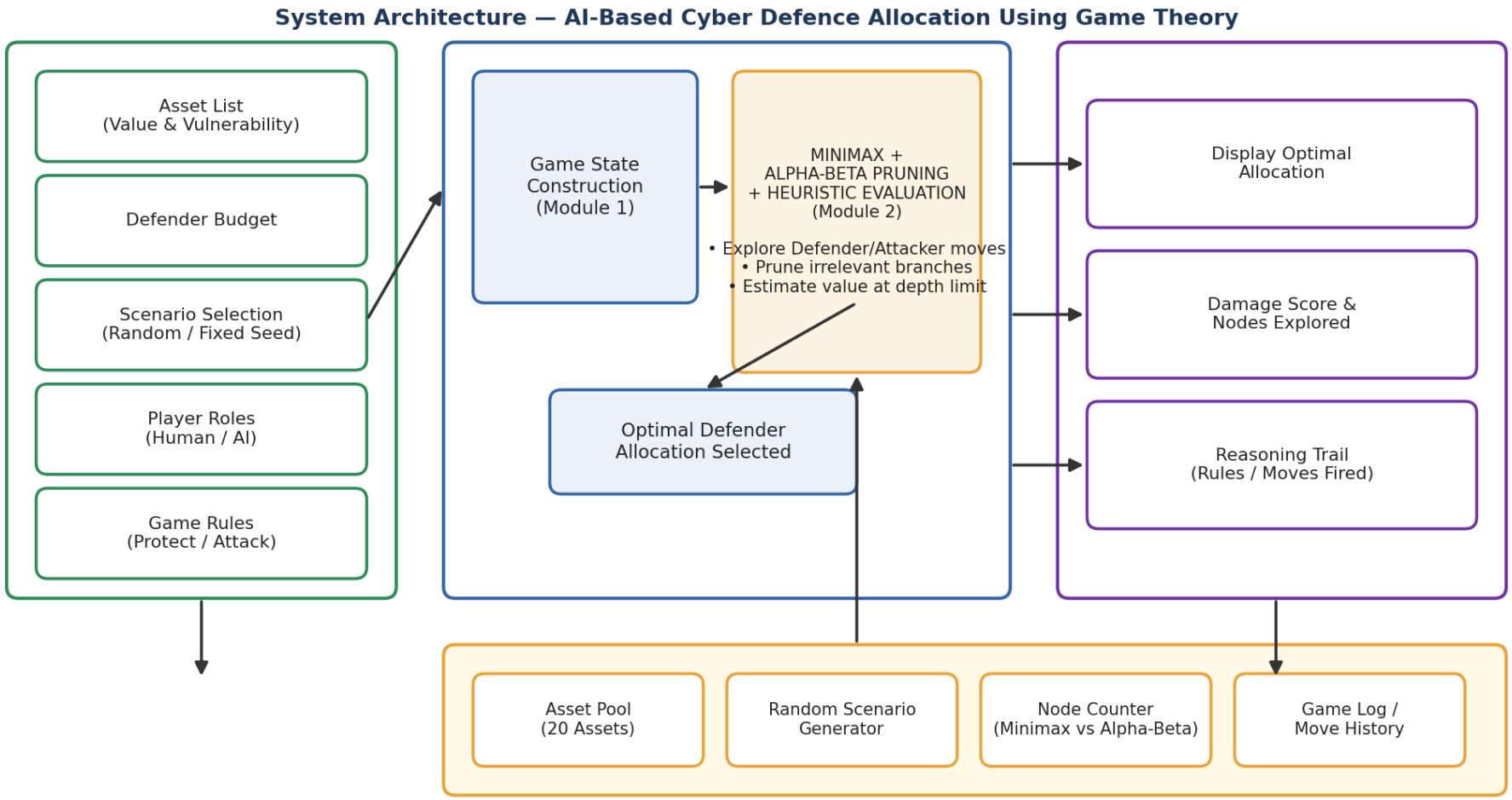
Working:

- Define the set of digital assets, their value and vulnerability.
- Set the Defender's limited protection budget.
- Defender allocates protection to assets within budget (Module 1).
- Minimax explores all possible Defender/Attacker move sequences.
- Alpha-Beta pruning skips branches that cannot affect the outcome.
- Heuristic evaluation estimates state value when a depth limit is reached.
- Optimal allocation and expected damage are displayed to the user.

Novelty in Proposed System

- **Dual-algorithm efficiency demonstration —** runs Minimax and Alpha-Beta on the same scenario to directly show pruning reduces nodes explored.
- **Scenario randomization —** a pool of 20 realistic assets, so every run features a different mix instead of one fixed case.
- **Flexible player roles —** supports Human-vs-AI (either role) or Human-vs-Human, not just AI-vs-AI.
- **Heuristic scalability —** a custom evaluation function lets the search handle larger asset counts without exhaustive search.
- **Explainable output —** a full reasoning trail (which move fired and why) instead of a black-box recommendation.

BLOCK DIAGRAM(PROPOSED SYSTEM)



Hardware/Software Requirements

Hardware Requirement:

- Processor : Intel Core i3 or above
- RAM 4 GB minimum
- Storage 1 GB
- Free Space Keyboard & Mouse

Software Requirement:

- Operating System : Windows 10/11
- Programming Language : Python 3
- IDE : Visual Studio Code
- Libraries : Tkinter (Python Standard Library only)
- Version Control : Git (Optional)

References

- **[1]** S. Russell and P. Norvig (2021). *Artificial Intelligence: A Modern Approach*, 4th ed. **Pearson.**
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- **[3]** S. Roy et al. (2010). *A Survey of Game Theory as Applied to Network Security*, Proc. 43rd Hawaii Int. Conf. on System Sciences (HICSS).
- **[4]** D. E. Knuth and R. W. Moore (1975). *An Analysis of Alpha-Beta Pruning*, Artificial Intelligence, vol. 6, no. 4, pp. 293–326.